



(Left to right) Dr Janani Venkatraman, Dr Santanu Datta, and Dr Arindam Ghatak, Founders, Biomoneta Research Pvt Ltd

ICUs to home care settings.

Dr Arindam notes that while specific electronic components were initially sourced internationally, the

company has transitioned to reliable Indian suppliers for most parts. Early challenges related to custom components and low-volume

requirements have eased as supplier ecosystems in Indian technology hubs have matured.

The company reports sales of over 750 ZeB-box-based units across India. Pricing varies by device type and application, ranging from approximately ₹20,000 for smaller units to around ₹250,000 for advanced medical devices.

Revenue is currently driven primarily by device sales, alongside ongoing efforts to expand awareness, strengthen distributor networks, and establish international partnerships.

The name Biomoneta reflects the company's core philosophy—bio representing life and health, and moneta, the Latin word for money.

—Nitisha Dubey

2 WHAT IF BATTERIES COULD BE MADE FROM FARM WASTE?

Twin sisters identified two converging challenges—rising pollution and the high cost of EV batteries—and responded by converting discarded crop waste into a safe, rechargeable battery chemistry. Could this mark a new direction for clean energy?

Nishita and Nikita Baliarsingh first discussed taking environmental action during a conversation on their terrace. Their research into electric vehicles (EVs) revealed batteries that were costly, reliant on scarce materials, and raised safety concerns. At the same time, pollution levels in North India were rising due to widespread crop-residue burning. This led to a simple but disruptive question, could agricultural waste be repurposed as battery material?

“During the 2020 lockdown, we ran a small at-home experiment using crop residue. To our surprise, it worked, producing nearly 5 volts. That result shaped our direction and led us to focus on building safe, clean batteries from natural bio-materials. We named the company Nexus to reflect connection, linking ideas, challenges, and battery cells into a single solution,” Nikita says.

The batteries are rechargeable and designed as direct replacements for lithium batteries. In principle, the technology could serve any application that relies on rechargeable batteries. For now, the primary focus remains EVs—including two-wheelers, three-wheelers, and four-wheel golf carts—as well as stationary applications such as battery energy storage systems (BESS). The company operates purely as a B2B business. “Our buyers include OEMs and companies developing energy storage facilities for solar and wind projects,” Nikita adds.

The core innovation lies in the battery manufacturing process. The batteries are produced entirely from bio-based nutrients and synthesised in-house, eliminating reliance on imported materials. In terms of performance, they are on par with lithium iron phosphate (LFP) cells in energy density, charge cycles,

voltage, and capacity. Although lithium nickel manganese cobalt oxide cells offer marginally higher performance metrics as a mature chemistry, these batteries remain safe, cost-effective, and competitive with LFP cells.

The journey involved several technical and operational challenges. One of the earliest and most critical hurdles was identifying suitable biomaterials. Although multiple candidate materials were evaluated, many failed to meet efficiency requirements. Technologically, the company has progressed significantly through sustained experimentation and analysis.

The startup operates a pilot manufacturing line in Bhubaneswar, which also serves as its headquarters. All current manufacturing is conducted on this pilot line. The process begins with material synthesis, followed by cell layering.